

PROBLEM_STATEMENT

🎯 Problem Statement & Alignment

The Problem

Hospitals, especially during high-demand periods (surges, emergencies, seasonal outbreaks), struggle to get a **real-time, unified view** of their available resources:

- Beds fill up across ICU, General, and Emergency wards without a central way to see occupancy at a glance.
- Doctor and nurse availability is tracked informally (whiteboards, phone calls, spreadsheets), making it hard to know who is free during a crisis.
- Critical equipment (ventilators, ECG machines, patient monitors) availability is not visible in one place.
- Administrators often notice a shortage only *after* it becomes critical, instead of being warned in advance.
- There's no simple way to simulate "what happens if patient load suddenly increases" to prepare staff and plan ahead.

Our Solution

The **Hospital Command Center & Resource Optimization** system directly addresses this by providing:

1. **A single input point** for hospital staff to enter current bed, staff, equipment, and patient numbers.
2. **Automatic calculation** of availability, occupancy percentage, and utilization — no manual math required.
3. **A rule-based recommendation engine** that flags shortages as NORMAL, WARNING, or CRITICAL before they become emergencies.
4. **A live dashboard** that updates instantly after any change, giving administrators a real-time command-center view.
5. **Simulation tools** (Normal Condition / Emergency Surge) so hospitals can stress-test their resource planning and train staff on crisis response.

Why This Matters

Without this system	With this system
Manual tracking, delayed awareness of shortages	Instant visibility into shortages as they emerge
No standard way to compare beds/staff/equipment together	One centralized dashboard for all resource types
Reactive decision-making	Proactive, alert-driven decision-making
Difficult to rehearse surge scenarios	Built-in surge simulation for planning and training

Alignment with the Goal

This project targets the core objective of **hospital resource optimization and decision support**: it doesn't just *display* data — it *ingests, processes, analyzes, and acts on* hospital data to surface actionable recommendations, which is the central ask of the problem statement.

Honest Scope Note

The current prototype uses **rule-based logic**, not machine learning, for its recommendations (e.g., “if occupancy $\geq$ 80% $\rightarrow$ WARNING”). This is intentional for a functioning, explainable, hackathon-stage prototype. It is designed to be extended with ML-based demand forecasting and predictive resource allocation — see [DEVELOPMENT_HISTORY.md](#) for the roadmap.